

EXECUTIVE SUMMARY

Anti-Money Laundering (AML) Detection Pipeline

Prepared by: Abdullah Hasan Mohamed | ID: 20221104 | May 2026

PURPOSE AT A GLANCE

This project delivers a fully automated machine-learning pipeline that identifies suspicious financial transactions indicative of money laundering — before they cause regulatory or reputational harm. Using publicly available synthetic bank data (PaySim), the system moves from raw transaction records to actionable risk scores through five structured phases, each designed for auditability, accuracy, and operational continuity.

Why This Matters

Money laundering costs the global economy an estimated \$800 billion to \$2 trillion annually. Regulators impose heavy fines and reputational damage on financial institutions that fail to detect it. Traditional rule-based systems catch only obvious patterns; sophisticated laundering schemes exploit their rigidity. A machine-learning approach continuously learns evolving behaviour, improving detection without relying on manually written rules.

Key business benefits of this pipeline:

Benefit	What it means for the business
Higher Detection Rate	ML catches subtle, multi-step laundering patterns that rule-based filters miss entirely.
Reduced False Positives	Fewer legitimate customers flagged incorrectly — less friction, lower investigation costs.
Regulatory Readiness	Transparent, explainable decisions and a documented audit trail support compliance reporting.
Continuous Improvement	The model updates itself as new transactions arrive — no costly full retraining required.

How the Pipeline Works — Five Phases

The system processes transactions sequentially through five interconnected stages. Each stage feeds the next, building a richer picture of risk with every step.

#	Phase	What happens	Business relevance
1	Data Preparation & Feature Engineering	Raw bank transaction records are cleaned, enriched with derived signals (e.g. whether a sender drained their account to zero), and formatted for analysis.	<i>Garbage in, garbage out. Careful preparation prevents the model from learning irrelevant patterns and ensures every useful signal is captured.</i>
2	Dimensionality Reduction (PCA)	Highly correlated financial variables are compressed into a smaller set of independent summary components, retaining 90%+ of the information.	<i>Reduces noise and speeds up processing — the model focuses on the most informative signals rather than redundant data.</i>
3	Behavioural Segmentation (Clustering)	Transactions are automatically grouped into distinct behavioural clusters — e.g. normal retail payments vs. large-value transfers between dormant accounts.	<i>Creates a contextual fingerprint for each transaction. Suspicious activity stands out more clearly when compared within its peer group.</i>
4	Risk Scoring (XGBoost)	A gradient-boosted model learns from labelled fraud examples to assign every transaction a probability of being fraudulent, using all engineered and cluster features.	<i>Delivers a precise, auditable risk score per transaction. Investigators can prioritise the highest-risk cases and justify decisions to regulators.</i>
5	Continuous / Batch Learning	As new transactions arrive each month, the model updates itself incrementally — no full retraining or access to historical data needed.	<i>Adapts to evolving criminal tactics automatically. Operationally efficient and safe: historical data never needs to be reloaded or re-exposed.</i>

Key Architectural Decisions — Plain Language

Several design choices were made deliberately to protect the integrity of the results and ensure the model performs reliably in production:

Decision	Why it was made
No "fraud flag" variable in training	The original dataset included a simulator-generated fraud flag. Including it would be cheating — the model would simply read the answer rather than learn the patterns. It was removed.
Measuring recall and F1, not accuracy	Fewer than 1 in 1,000 transactions are fraudulent. A model that flags nothing would be 99.9% accurate but useless. Recall (how many frauds are caught) and F1 (balance between catch rate and false alarms) are the correct measures.
Handling class imbalance directly	XGBoost was configured to weight fraudulent transactions proportionally higher, preventing the model from being overwhelmed by the sea of legitimate transactions.
Separate scalars for training and live data	The model was deliberately prevented from seeing any test-set statistics during training. This ensures performance figures reflect real-world accuracy, not an optimistic lab result.
Warm-start incremental learning	Monthly model updates add only new decision trees on top of the existing model. Historical data is never reloaded — reducing privacy risk and infrastructure cost.

Conclusion & Recommended Next Steps

This pipeline demonstrates that a structured, five-phase machine-learning approach can detect money laundering with high reliability — while remaining explainable, adaptable, and operationally sustainable. The system is designed to grow alongside the institution's transaction volumes and to keep pace with evolving criminal tactics through its monthly update mechanism.

Recommended next steps for stakeholder review:

1. Validate model performance on real institutional transaction data (rather than synthetic PaySim data) with the compliance team.
2. Define the operational alert threshold in collaboration with the fraud investigations unit — balancing investigator workload against missed-fraud risk.
3. Establish a data governance framework for the monthly batch-update process, including sign-off procedures and model versioning.
4. Commission an independent model validation review as required by SR 11-7 model risk management guidelines.